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Privacy-preserving data mining for open government data from heterogeneous sources

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ABSTRACT

Open data is a global movement with the potential to generate significant social and economic benefits. Policies on open government data (OGD) inspire the development of new and innovative services that government agencies may lack. The International Open Data Charter adequately describes the importance of data mining. Governments that have signed this charter should focus on the following areas—(i) data mining, (ii) linkage, and (iii) in-depth analysis, i.e., distribution of open data that is freely accessible for elaborate analysis using machine reading. However, a series of practical difficulties is observed in connection with the data mining of OGD for in-depth analysis. First, most OGD do not have identifiers to prevent privacy disclosure. Second, owing to the nature of siloed data, the data sharing and collection methods vary with respect to heterogeneous OGD, and administrative or institutional barriers need to be overcome. This has created a demand for a novel technical solution that applies micro-aggregation and distance-based record linkage to address the aforementioned issues. Thus, in this study, a method capable of integrating two or more de-identified OGDs into one dataset to enable OGD data mining is proposed. In addition, the proposed method allows users to adjust the privacy threshold level to determine an appropriate balance between privacy disclosure risk and data utility. The effectiveness of the method is evaluated in terms of several metrics via extensive experimentation. This study emphasizes the importance of the research on efficient utilization of already-published OGDs, which has been relatively neglected in the past. Further, it broadens the research area for privacy-preserving data mining by proposing a method capable of mining heterogeneous data even in the absence of identifiers.

1. Introduction

In this study, we propose a micro-type privacy-preserving data mining (PPDM) algorithm to protect the privacy of open government data (OGD) without any identifiers during mining.

In 2013, several nations signed the G8 Charter. However, the possibility of refining its principles to support wider utilization of OGD has since been discussed (The International Open Data Charter, 2020). The preamble to the International Open Data Charter clearly highlights the primary purpose of OGD to be the pursuit of technology-fueled innovation to create more accountable, efficient, responsive, and effective governments and businesses, while simultaneously promoting economic growth (G8, 2013).

The public use of open data is necessary to achieve the aforementioned goals (T. M. Yang & Wu, 2016). However, this grants stakeholders across the world unprecedented access to OGD, enabling outsiders to use their external experience and information gathered from data mining as leverage for their own interests (Altayar, 2018; G8, 2013; Janssen, Charalabidis, & Zuiderwijk, 2012; Z. Yang & Kankanhalli, 2013; Zuiderwijk, Helbig, Gil-García, & Janssen, 2014).

Data mining is a process that uses sophisticated data analysis tools to investigate relationships present in datasets, revealing hidden patterns that are otherwise indiscernible (Tan, Steinbach, & Kumar, 2016). It includes statistical modeling techniques, mathematical algorithms, and various machine learning-based methods (Seifert, 2004).

Expanding the scope of OGD implementations requires the implicit use of heterogeneous data sources (Conradie & Choenni, 2014). However, the utilization of data from various sources is difficult owing to the de-identification of personal information present in OGD data. Such privacy barriers present a major hindrance to OGD mining by restricting access to sensitive data (Axelsson & Schroeder, 2009; Ruijer, Grimmelikhuijsen, & Meijer, 2017). Additionally, implementations can be

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significantly delayed owing to the time-consuming and expensive nature of legal proceedings associated with the participating services (Conradie & Choenni, 2014), which are often geographically separated. Some practitioners have suggested that a technical solution might be a more appropriate solution to the privacy issue, to enable special measures that can be implemented in the workplace immediately (Seifert, 2004).

In summary, OGD mining is difficult. This is because of the complexity of establishing good linkages for data analysis owing to the lack of identifiers caused by de-identification. Silos further complicate data-sharing between government agencies. Therefore, in this study, we formulated a research question on the possible modalities of application of the PPDM method to OGD without using identifiers. To answer this question, we propose a new technical solution to link heterogeneous data via a privacy-preserving technique.

PPDM of big data based on artificial intelligence (AI) is a state-of-the-art technical solution that preserves sensitive attributes. Consequently, it has become an increasingly popular field of research (Agrawal & Srikant, 2000). PPDM-related studies primarily aim to develop an efficient algorithm capable of preventing disclosure or inference of sensitive data during the extraction of relevant information from databases (Wu, Chu, Wang, Liu, & Yue, 2007).

PPDM finds applications in several practical scenarios, as demonstrated by its role in medical research based on personal medical records of patients (Lindell & Pinkas, 2002). For example, the Center for Disease Control researches patterns of disease occurrence by analyzing the data owned by various insurance companies. However, if an insurance company were to disclose the entirety of personal medical data to the Center for Disease Control, it would create serious commercial and legal problems (Estivill-castro & Clifton, 2002). In such a case, a distributed data mining algorithm is required to safeguard the private information of the associated users (Xiong, Chitti, & Liu, 2007).

However, almost all existing studies related to the adoption of privacy-preserving solutions with respect to OGD have focused on technical solutions during the publishing stage, restricting their application to future OGD deployments. Application of such methods to previously distributed data would be considerably costly as the data would first have to be relocated. This makes it apparent that a solution that allows the utilization of OGD that have already been anonymously distributed is urgently required.

The proposed method is a hybrid model, combining distance-based record linkage with a micro-aggregation method. It data mines de-identified OGD via record linkage. Thus, the method is capable of addressing the problem of mining of anonymized and already distributed OGD. Further, it supports heterogeneous data mining for in-depth analysis, which has been emphasized in the Open Data Charter. The rest of this study is organized as follows.

The fundamental materials and methods associated with this study and the related works are discussed in Section 2. The pseudo-codes for the algorithm are outlined in Section 3. The experimental environment and the results are summarized in Section 4. The results are contextualized and discussed in Section 5. Finally, the study is concluded in Section 6.

2. Materials and methods

2.1. OGD policy backgrounds and the status of privacy protection systems in major countries

Numerous existing empirical studies have documented the benefits of adopting OGD (Altayar, 2018). According to reviews, it can help to improve transparency and accountability of government agencies and the developmental process of policies and procedures. Publication of OGD supports the development and deployment of services, products, and applications that help both societies and governments to enable external stakeholders to access, investigate, discover, and process government data (Altayar, 2018). However, it also renders OGD susceptible to being used as leverage by people for their own interests (Z. Yang & Kankanhalli, 2013; Zuiderwijk et al., 2014). Therefore, adequate measures are required to protect government resources from potential security threats (Zhou & Hu, 2008).

Countries across the world have implemented various legislative systems to de-identify personal information and enable the effective utilization of OGD. Further, each country uses its own code of conduct. In the European Union (EU), de-identified personal information is known as anonymous information. Additionally, the Article 29 Working Party (WP29) outlines the anonymization techniques that must be performed on open data; the EU emphasizes the adoption of adequate safeguards for utilization of OGD in scientific, historical, and statistical research in Article 6.

In the United States, the term “de-identification” is officially used to signify the concept of de-identified information. According to the Family Educational Right and Privacy Act (FERPA) and Health Insurance Portability and Accountability Act (HIPAA), the criteria for de-identification are independently defined for each field of application, such as medicine and education. Further, the NISTIR 8053 document published in 2015 also emphasizes the revision and supplementation of institutional frameworks for personal information and the promotion of data activation policies.

Finally, in Korea, information that has been de-identified is simply referred to as de-identified information. Via the guidelines for the definition and de-identification of personal information published in 2016, the scope of utilization of personal information was explicated along with the legal measures associated with the de-identifying processes. According to Article 18 Section 2 of the Personal Information Protection Act of Korea, the use of personal information for purposes other than the intended ones and their provisional use by third parties are partially permitted. Such third-party exceptions apply to statistical analysis and academic research. Additionally, information is considered to be de-identified when personal information is provided in a form in which specific individuals cannot be identified (Kim, 2017). The aforementioned features have been summarized in the appendix (Table A).

2.2. Current trends in OGD privacy protection techniques

Certain privacy advocates argue for clearer policies and stronger oversight, emphasizing the establishment of strong privacy laws. On the contrary, other practitioners suggest that a technical solution for privacy-related issues is more appropriate. They argue for the urgent adoption of measures that can be applied in the workplace immediately (Seifert, 2004).

According to Thompson, Mullins, and Chongsutakawewong (2020), de-identification techniques, which comprise one such set of technical solutions, remove identification information from datasets while retaining the rest of the utilities of the data (Ceross & Simpson, 2018). It is hardly a secure task because privacy and security considerations always pose moral and legal barriers to the distribution of datasets (Janssen & van den Hoven, 2015).

The concepts, processing targets, and advantages and disadvantages of various de-identification techniques have been summarized in Table B in the appendix. However, in practice, even if sufficient de-identification measures have been executed via the aforementioned techniques, data can still be regarded as personal information if it does not pass the adequacy assessments associated with anonymization. This is owing to the possibility of inference via reverse engineering by combining the data with additional information, which could result in the identification of specific individuals even in the absence of identifiers. In general, the following anonymization tests are primarily used to assess the adequacy of the de-identification process—k-anonymity, l-diversity, and t-closeness.

k-Anonymity is a technique that simply examines whether a specific individual can be re-identified from the available data. This method is one of the most basic assessment techniques and is easy to understand and execute. However, it has the drawback of being vulnerable to several types of attacks, including homogeneity attacks and background knowledge attacks (Machanavajjhala, Gehrke, & Kifer, 2006).

l-Diversity assigns well-represented sensitive values greater than or equal to l to each equivalence class. This addresses some of the disadvantages of k-anonymity. It extends k-anonymity by additionally requiring the existence of well-represented values within each anonymity group. However, *l*-diversity fails to prevent probabilistic inference attacks and attribute disclosure (Machanavajjhala et al., 2006).

t-Closeness is a further extension of *l*-diversity. Instead of merely guaranteeing the well-representation of sensitive values, this approach requires the distribution of every sensitive attribute inside the anonymity groups to be identical to the distribution of the attribute over the entire dataset, modulo a threshold t. It protects against attribute disclosure, but not against identity disclosure (Ninghui, Tiancheng, & Venkatasubramanian, 2007).

In general, a higher level of de-identification leads to lower risk of privacy breaches, but also lower data utility. Conversely, a lower level of de-identification leads to higher risk of privacy breaches, while providing greater data utility. Therefore, an appropriate level of de-identification measure control is required.

2.3. Another issue with OGD data mining

In addition to issues related to privacy, utilization of OGD faces several other challenges that need to be overcome. Silos present one such problem. The government is the producer as well as the consumer of the data. Following digitization, the vast amount of OGD collected from various sources, such as tax systems, social programs, health records, etc., can be used for decision-making in the fields of education, economics, health and social policy. However, several government departments utilize the siloed data structure, which engenders institutional issues during the processing of vast amounts of OGD due to bureaucrats (Giest, 2017). The siloed data is equipped with legal components that prevent indiscriminate sharing of data based on privacy laws. Therefore, these legal components, along with the institutional environment such as bureaucrats, serve as major obstacles to the data-sharing and collection methods (Giest, 2017). Although the public silo system used to be primarily related to institutional structures, it now includes information technology (IT) and data elements as well (Bannister, 2001, 2015). An IT silo system (also known as a stovepipe system) refers to a system developed to reduce complexity and create clear rules for reporting and decision-making. However, the silo system has become an obstacle in the policy-making process owing to increasing cooperation and interdepartmental issues (Giest, 2017).

Owing to the existence of such silos, a greater emphasis has been laid on integrated platform-based governance and e-government platforms in recent years (Huber, Kude, & Dibbern, 2017; Janssen & Estevez, 2013; Schwester, 2009; Wimmer et al., 2009).

The transaction costs incurred during the data mining of siloed databases corresponding to exclusive government departments can present significant challenges to the implementation of OGD infrastructures. Further, wider implementations implicitly require the data mining of heterogeneous databases, which is also computationally expensive (Conradie & Choenni, 2014). Moreover, there are additional difficulties arising from associated administrative processes or judicial enactments and the standardization of databases.

The utilization of linked data enhances the usability of open data by reducing the burden of such preparatory processes. Links allow facile search for relevant datasets and the adjustment of multiple datasets. They also make it easier to provide data users with proof information, attention to specific data values, definition of terms, and version information. As such, the use of linked data facilitates open data agendas (Shadbolt & O'Hara, 2013). However, an adequate amount of linked data is currently not available. For instance, in the United Kingdom (UK), less than 5% of the 9000 datasets available at data.gov.uk contain linked data. According to the linked data cloud (http://road-cloud .net/) of the Linked Open Data project, urgent measures are required to link OGD as approximately one-sixth of the contents of Linked Data Warehouse (LDW) are not linked (Shadbolt & O'Hara, 2013).

3. Theory and calculations

3.1. Data and variable characteristics

The databases used in the experiments presented in this study were primarily obtained from two sources—the Korean Innovation Survey (KIS) and the Korea Enterprise Data (KED). Each database is a multi-year micro-dataset obtained from the relational database management system (RDBMS) for separate purposes. The database acquired from KIS comprises cross-sectional survey data obtained via an annual survey on the research and development (R&D) activities of Korean companies. This survey captures contemporary international trends by following the Oslo Manual (OECD, 2005b) of the Organization for Economic Co-operation and Development (OECD), which facilitates international comparison. It acquires and provides basic data required for the establishment of national innovation policies and innovation research by identifying the current status and characteristics of all innovation-related activities of manufacturing and service industries.

Conversely, the KIS database comprises various general details regarding companies, including company history, employee information, sales figures, expenditure, patents, anchor product ratios, and technological levels. In addition, information on R&D investments, equipment purchases, and government-supported programs sorted by types of innovation activities (product, process, organization, and marketing) is also included. The KED database comprises longitudinal data containing the financial information of Korean companies, including the data disclosed in financial position statements and cash flow statements obtained from the Korea Corporate Disclosure System (Jun, Lee, & Lee, 2020). The linking of the two aforementioned databases would enable the investigation of a host of topics, like variations in the time series of a company's financial performance with respect to its R&D activities.

4000 companies from the manufacturing sector that responded to the survey between January 1, 2013 and December 31, 2015 were used as the analysis targets in this study from the KIS database (certain items, however, corresponded to the period between January 1, 2015 and December 31, 2015). The analysis targets from the KED dataset comprised the financial information of 119,890 companies surveyed in 2015. Throughout this study, the data of 4000 companies surveyed in 2015 obtained from the KIS database is referred to as the basic database, while the data of 119,890 companies collected in 2015 obtained from the KED database is referred to as the extended database. We acquired the business registration numbers of 1055 companies that appear in both databases. The linkage proposed in the method developed in this study is executed without the knowledge of any identifiers, including the business registration numbers that we obtained. Instead, the business registration numbers for 1055 companies are used only during the performance assessment of the proposed linkage technique.

The variables used in the experiments are industry category, establishment year, and sales figures, which are common to both databases. In analogy with personal data, industry category can be compared to the residential zip code of an individual, while the establishment year and sales figures can be regarded to correspond to his age and income, respectively. Industry category and establishment year are considered to be quasi-identifiers (QI). These two variables do not uniquely identify particular companies by themselves, but they can be used in conjunction with other information to identify them (OECD, 2005a). Additionally, the data used in the experiments in this study are secondary data that were distributed in accordance with the strict guidelines for privacy protection established by the OECD.

3.2. Synthetic data generator for PPDM

Synthetic data-based techniques have garnered the interest of various public institutions and several attempts have been made to apply such techniques to OGD. The SIPP Synthetic Beta (SSB) dataset is a synthetic dataset produced by the US Census Bureau that integrates Survey of Income and Program Participation (SIPP) micro-data with tax-related micro-data obtained from the Social Security Administration (SSA) and the Internal Revenue Service (IRS) (Abowd, Stinson, & Benedetto, 2006; Benedetto, Stinson, & Abowd, 2018). Moreover, other attempts have been made a Longitudinal Business Database to integrate US Census Bureau data with the Internal Revenue Service data (Kinney et al., 2011). In Germany, experiments related to employment statistics have been conducted to integrate the German Social Security Data with the IAB Establishment Panel data as well (Drechsler, Bender, & Rässler, 2008).

This study focuses on the method introduced by Domingo-Ferrer and González-Nicolás (2010) that applies micro-aggregation to synthetic data generators to link and utilize heterogeneous OGDs (micro-data). Synthetic data generation refers to the complete transformation of data based on multiple imputation (Rubin, 1993). Domingo-Ferrer and González-Nicolás (2010) identified that even though micro-aggregation methods are capable of preserving broad distribution characteristics of almost all records, they accurately preserve them in only a few cases, if at all in any. Conversely, although synthetic data generation preserves pre-collected statistics accurately and discloses a smaller amount of personal information compared to micro-aggregation, it suffers from the limitation that statistics other than pre-collected statistics are not preserved at all. Therefore, the authors proposed a combination of micro-aggregation and synthetic data generation. They created a group comprising at least k (usually 3) records from the original data and then synthesized a new value based on it via multiple imputation while preserving the statistics of the data values belonging to each group (Domingo-Ferrer & González-Nicolás, 2010). However, the application of this method for de-identification of data is difficult and its significant impact on data utility and disclosure risk caused by the selection of the synthetic data generator cannot be controlled. Moreover, if the performance of an imputation model is very good, the generated synthetic data is liable to be too similar to the original data. Therefore, the selection of the imputation model greatly affects subsequent data utility (i.e., which distribution is preserved) and disclosure risk (Domingo-Ferrer & González-Nicolás, 2010; Reiter, 2003, 2005).

In addition, the multiple imputation method employs simulations and adds random noise to the data, rendering the imputation of new values unacceptable at the initial value of data (Patrician, 2002; Rubin, 1996). Owing to the incorporation of randomness into the analysis, each calculated estimate corresponding to each imputed dataset differs slightly from the others, and identical results cannot be acquired (Patrician, 2002).

3.3. Proposed method

The US Census Bureau created an integrated dataset by generating synthetic data based on SIPP micro-data, SSA, and IRS micro-data as identifiers.

This is a viable method in this case because the US Census Bureau holds the identifier information. However, most companies, research institutes, and universities do not have an equivalent right to access such identifiers. To address this shortcoming, a PPDM method that can be used to de-identify data is proposed in this study. First of all, the record linkage of heterogeneous data is performed based on the calculated distances between the common tuples of the de-identified data. In addition, the values similar to the previously collected statistics are preserved (synthetic data) by categorizing the linked data into groups comprising at least k (usually 3) records in the same order as in the original data. Finally, the data values belonging to each group are replaced by the average value of that group (micro-aggregation). The fundamental feature of the proposed method is that the number of record linkages can be adjusted by appropriately altering the privacy threshold level. If a lot of records similar to the original data are observed to be linked, the privacy level can be raised by lowering the disclosure risk, decreasing the probability of re-identification of records that share their identifiers with the original data. However, owing to the overabundance of similar records, the amount of noise is high and data utility is low. Conversely, if a low number of records similar to the original data are linked, the privacy level can be lowered, increasing the risk of re-identifying records that share their identifiers with the original data. However, because the number of similar records is small in this case, the noise is low and the utility is relatively high. The former case exhibits high privacy and secures a high number of similar records, and is thus suitable for marketing surveys that wish to identify policy targets. Conversely, the latter case is suitable for research because it exhibits high utility and a low amount of similar records. This discussion is summarized in Fig. 1.

3.4. Process of OGD (with no identifiers) mining using the proposed method

The SSB produced by the US Census Bureau is a useful example to explain the utility of the proposed model. Although generally companies or institutions do not possess the authority to search for identifiers (resident registration number or business registration number) shared between the original and heterogeneous data like the US Census Bureau, the method proposed in this study can be used to construct a dataset comprising an integration of the aforementioned data, as depicted in Fig. 2. In SSB, the SIPP serves as the raw data and the SSA and IRS comprise the heterogeneous data. In the proposed model, the original data comprise the basic DB (database), and the heterogeneous data comprise the extended DB. Therefore, SIPP is denoted by DB_A, and SSA and IRS are denoted by DB_B and DB_C, respectively. DB_A, DB_B, and DB_C share a common tuple (quasi-identification information). For example, if the common tuple is (x,y,z), $DB_A = (x,y,z,A)$, $DB_B = (x,y,z,B)$, and $DB_C = (x,y,z,C)$. The proposed model performs record linkage between original data and heterogeneous data using the common tuples (x,y,z), and generates synthetic tuples of linked heterogeneous data. Finally, DB_{ABC} is constructed by integrating the generated synthetic tuples into the original data. Thus, the integrated DB_{ABC} is composed of the tuples, $\left(x,y,z,A,\tilde{B},\tilde{C}\right)$

3.5. Expected objectives and evaluation methods of the study

This study empirically evaluates the performance of the proposed method in de-identifying OGD by using KIS as the basic DB, representing the original data, and KED as the extended DB, representing heterogeneous data to be linked to the original data. This study was conducted under the assumption that the business registration number, which is an identifier shared between the KIS and KED, is completely unknown. The business registration number is used only during performance evaluation. The proposed method only uses the common tuples of continuous type shared by the basic DB (KIS) and the extended DB (KED). At this time, categorical tuples can be processed by filtering only those cases in the extended DB (KED) whose categorical tuple values in the extended DB (KED) are identical to the categorical tuple values in the basic DB (KIS).

In other words, the distance for the linkage of records is calculated based on tuples of continuous type in the actual model and the model for the categorical tuples functions by excluding the cases with categorical tuple values that do not match the basic DB (KIS) in the extended DB (KED) based on the target calculated distance. In each database, the constituent data are expressed as tuples of the form (x,y,z), where x denotes the industry category, y denotes the establishment year, and z

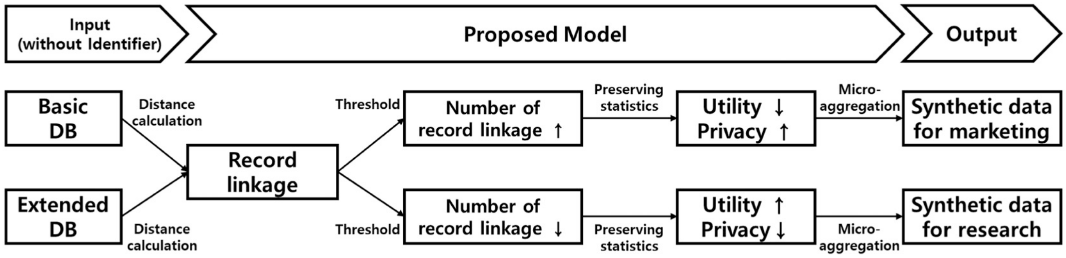

Fig. 1. Synthetic data generation based on the purpose of use.

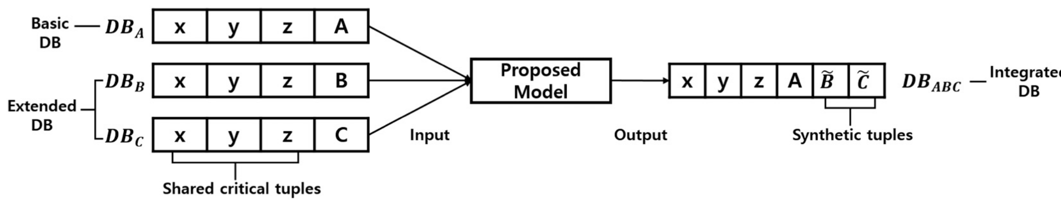

Fig. 2. Example of building integrated data through the proposed model.

denotes the sales figure of a particular company. In each database, the establishment year is considered to be a QI, and the sales figure is considered to be a sensitive attribute (SA).

The evaluation method estimates the effectiveness of the generation of optimal synthetic data, in which data utility and privacy share a balanced, trade-off relationship, based on the threshold control function. The fulfillment of the objective of striking a balance between utility and privacy during the linkage of OGD without identifiers can be assessed based on multiple metrics—the degree of OGD linkage in the absence of identifiers, the utility of the linked results, and the degree to which privacy is maintained. The details of the evaluation are as follows. As discussed previously in Section 3.3, the number of linked records relative to the original data greatly affects data utility and privacy. Thus, the numbers of record linkages achieved based on two separate distances—Euclidean distance and Mahalanobis distance—were compared in this study. Then, verification was conducted to assess whether records with identical business registration numbers had been included in identical record groups linked to the original data. This also represented the effectiveness of the proposed method in creating groups comprising at least k (usually 3) records from the heterogeneous data that are similar to the original data in the absence of any identifiers. Therefore, this evaluation result represents data utility in this study. In particular, this study aims to set business registration numbers as labels and measure the hit rate of the proposed model.

It estimates the relationship between results predicted by the model and their actual counterparts. When the model predicts an actually True case to be True, the result is considered to be a True Positive (TP). Similarly, when it predicts an actually False case to be True, the result is considered to be a False Positive (FP); when it predicts an actually True case to be False, the result is considered to be a False Negative (FN); and when it predicts an actual False case to be False, the result is considered to be a True Negative (TN). During this evaluation process, the hit rate is defined to be the rate at which the model predicts actually True cases to be True, i.e., Hit rate = TP / TP + FN. If a True label is assigned to a linked candidate group derived by the proposed model, it is considered to be a hit.

Finally, the proposed method replaces the data values of similar records belonging to each group with the average value of the group and converts it into a new value to evaluate the difference between the privacy levels of the generated and original data. To this end, a *k*-anonymity test was performed to measure the level of privacy in this study. *k*-anonymity is required in order to prevent re-identification of de-identified personal information. Owing to these characteristics, although k-anonymity is basically a privacy protection model proposed to protect against vulnerabilities to linkage attacks on data, it was also used to evaluate the adequacy of de-identification measures (Sweeney, 2002). The difference in the privacy level was evaluated using the RMSE (root mean square) of the k-anonymity test result of the original data measured in this way and the k-anonymity test result of the synthesized data generated by the proposed method. In addition, a matching-sample *t*-test was performed on the probability of re-identification of individual records of the k-anonymity test result of the original data and the k-anonymity test result of the synthesized data generated by the proposed method to statistically verify if there is a significant difference in RMSE.

4. Results

4.1. Developmental results of the proposed PPDM linkage method

4.1.1. Pseudo-code for the synthetic data generator for PPDM

The algorithms developed in this study can be divided into the following five categories: (1) a method to calculate distances between records in the basic database, (2) a method to derive the optimal classifying threshold based on the estimated distances, (3) a method to calculate the distances between records in the extended database and those in the basic database, (4) a method to determine successful linkages between pairs of records from the basic and extended databases based on the aforementioned threshold, and (5) a method to conduct micro-aggregation of records in the extended database that have been successfully linked to records in the basic database. Algorithms (1–4) perform record linkage and Algorithm (5) addresses the micro-aggregation of similar linked records and finally converts them to new values similar to the original data.

Algorithm 1 calculates the distance matrix corresponding to the basic database based on only the continuous variables. In the algorithm, $dist_{basic}$ denotes the matrix whose (*i, j*)th entry corresponds to the distance between the *i*th and *j*th records of the basic database.

This distance matrix is a (upper or lower) triangular matrix, and it is strictly triangular because all entries on the major diagonal are 0. Therefore, $n(n-1)/2$ calculations are required to estimate all entries of the distance matrix. Here, the distance can be taken to be either the Euclidean distance or Mahalanobis distance. Consider two points—$P = (p_1, p_2, p_3, \ldots, p_n)$ and $Q = (q_1, q_2, q_3, \ldots, q_n)$. The Euclidean distance between *P* and *Q* is given by Eq. (I.1).

$$D_{euclidean}(p,q) = \sqrt{\sum_{i=1}^{n}(p_i - q_i)^2} \tag{I.1}$$

Conversely, the Mahalanobis distance considers the correlations between variables, and is measured using a covariance matrix. Thus, the distance can be calculated by considering variance. It is given by Eq. (I.2).

$$D_{mahalanobis}(p,q) = \sqrt{(p_i - q_i)\sum^{-1}(p_i - q_i)^T} \tag{I.2}$$

Algorithm 1 Distance matrix of basic database

Algorithm 2 outlines the method of deriving an appropriate threshold based on the distance matrix of the basic database. $dist_{basic}$ may be considered to denote a distance matrix of length n(n − 1)/2. The entries of $dist_{basic}$ are first arranged in descending order. Then, n(n − 1)/2, which denotes the length of the distance entries corresponding to $dist_{basic}$, is multiplied by the desired confidence level, e.g., 0.95, 0.99, or 0.999, to derive the *index* value, *m*. Finally, the *m*th entry in the sorted list of entries of $dist_{basic}$ is selected to be the threshold.

Algorithm 2 Calculation of threshold based on the distance matrix of the basic database

An illustrative application of Algorithm 2 is depicted in Fig. 3. Suppose that the distance matrix of a basic database with 30 records is calculated using Algorithm 1, as depicted in the upper table of Fig. 3. The number of non-zero entries in this matrix is 435 (calculated using the formula, n(n − 1)/2). Subsequently, all non-zero entries of the matrix are arranged in descending order, as illustrated in the two tables in the bottom left of Fig. 3. Then, the rank of each value in the sequence is calculated, divided by the length of the sequence, and approximated using 0.95, 0.99, or 0.999 to obtain the final threshold. To simplify the algorithm, its pseudo-code inverts the rank by directly multiplying the length of the descending sequence with 0.95, 0.99, or 0.999 to calculate the threshold value. The aforementioned implementation is instantiated in the image on the right of Fig. 3.

Algorithm 3 outlines the method of deriving the distances between records in the extended database and those in the basic database. Let $dist_{extended}$ denote the matrix whose (*i, j*)th entry corresponds to the distance between the *i*th record of the basic database and the *j*th record of the extended database. Thus, in order to calculate $dist_{extended}$, corresponding to a basic database comprising *n* records and an extended database comprising *m* records, $n \times m$ separate operations are necessary. The distance matrix, $dist_{extended}$, is derived following a procedure similar to the method of deriving $dist_{basic}$, as outlined in Algorithm 1.

Algorithm 3 Distance matrix corresponding to a basic database and an extended database

Algorithm 4 outlines the method to determine successful linkages between entries of the basic database and those of the extended database based on the pre-established threshold. For this purpose, the distance matrix of the extended database and the pre-calculated threshold are used. If the distance between one record of the extended database and one record of the basic database is lower than or equal to the value of the threshold, the pair is considered to be successfully linked. Otherwise, it is not considered to be successfully linked; it is also referred to as a failed linkage.

Algorithm 4 Determination of success or failure of linkage

Algorithm 5 outlines a method to conduct micro-aggregation of the records of the extended database that have been successfully linked to some records of the basic database. The record linkage list comprises records belonging to the extended database that fulfill the given threshold corresponding to each record belonging to the basic database. The minimum number of records required in the record list (at least 3) serves as an additional hyper-parameter during this step. Based on this number, the proposed model performs micro-aggregation, which calculates appropriate averages of the records belonging to the extended database. Subsequently, the average values are used to constitute new synthetic data for sensitive information in the basic database.

Algorithm 5 Micro-aggregation to generate synthetic data

Algorithm 1 Distance matrix of basic database

Input: A basic database, DB_{basic}, that shares critical tuples with the extended database for record linkage

Output : The distance matrix of the basic database, $dist_{basic}$

1: **for all** $i = 1$ to n **do** // n is equal to the number of rows in DB_{basic}

2: **for all** $j = 1$ to n **do**

3: **if** $dist_{basic}$ is taken to be Euclidean distance **then**

4: **return** $dist_{basic}$=$D_{euclidean}(DB_{basic}[i], DB_{basic}[j])$

5: **else** $dist_{base}$ is taken to be Mahalanobis distance **then**

6: **return** $dist_{basic}$=$D_{mahalanobis}(DB_{basic}[i], DB_{basic}[j])$

7: **end if**

8: **end for**

9: **end for**

10: **return** $dist_{basic}$

Algorithm 2 Calculation of threshold based on the distance matrix of the basic database

Input: The confidence level, C, and the distance matrix of the basic database, $dist_{basic}$

Output: The threshold to be used during record linkage, T

1: **sort in descending power** distance entries of $dist_{basic}$

2: **if** T corresponds to 95 % **then**

3: **compute** $index_{95}$ equals the product of the length of entries of $dist_{basic}$ with the confidence level, C (0.95)

4: **return** T equals $index_{95}$-th entry in the sorted distance distribution of $dist_{basic}$

5: **else if** T corresponds to 99 % **then**

6: **compute** $index_{99}$ equals the product of the length of entries of $dist_{basic}$ with the confidence level, C (0.99)

7: **return** T equals $index_{99}$-th entry in the sorted distance distribution of $dist_{basic}$

8: **else** T corresponds to 99.9 % **then**

9: **compute** $index_{99.9}$ equals the product of the length of entries of $dist_{basic}$ with the confidence level, C (0.999)

10: **return** T equals $index_{99.9}$-th entry in the sorted distance distribution of $dist_{basic}$

11: **end if**

12: **return** T

4.2. Evaluation of the proposed model

4.2.1. Evaluation of the number of record linkages corresponding to each distance calculation method

The most fundamental hyperparameter is the minimum number of entries in the record linkage list. Although Euclidean distance exhibited relatively low linkage success rate in the basic database compared to Mahalanobis distance, its average number of entries in the record linkage list was higher. Further, with an increase in the confidence level in Algorithm 2, both distance metrics exhibited decrements in the corresponding linkage success rates in the basic database and the average number of linkages in the extended database.

4.2.2. Evaluation of data utility

Data utility was evaluated based on previously obtained business registration numbers. Corresponding to the Euclidean distance, the hit rates were observed to be 0.423 (Threshold = 95%), 0.402 (Threshold = 99%), and 0.297 (Threshold = 99.9%), whereas corresponding to the Mahalanobis distance, the hit rates were 0.485 (Threshold = 95%), 0.429 (Threshold = 99%), and 0.343 (Threshold = 99.9%).[1] Thus, Euclidean distance exhibited a lower hit rate compared to Mahalanobis distance. Combining these results, we conclude that the Euclidean

[1] A difference was identified between the units of measurements in the survey corresponding to the sales number attribute variable in the two databases. One of the databases used the unit, 1 KRW, while the other used the unit, 1,000,000 KRW, for the sales numbers. This exemplifies one of the issues with OGD mining as discussed previously in Sections 2 and 3. As compensation, the errors arising from different measurement methods between heterogeneous databases are considered during distance calculation. Meanwhile, if the threshold confidence levels are set to 50% and 75%, the hit rate can be significantly increased to 0.603 (EUC 50% = 0.482, EUC 75% = 0.457, MAH 50% = 0.603, MAH 75% = 0.568). However, we do not recommend using such thresholds as a high hit rate exhibits a trade-off relationship with the level of data privacy. Our primary aim is to achieve a balance between hit rate and privacy rather than maximizing one element at the cost of the other. Therefore, we determine a maximum hit rate of 0.485 to be sufficient.

Distance matrix of example basic database (number of records 30, length of entries 435)

	Record 1	Record 2	Record 3	…	Record 28	Record 29	Record 30
Record 1	0						
Record 2	0.185	0					
Record 3	0.702	0.573	0				
…	…	…	…	0			
Record 28	…	…	…	…	0		
Record 29	…	…	…	…	…	0	
Record 30	…	…	…	…	1.000	0.722	0

$dist_{basic}$

Index	Distance
1	0.185
2	0.702
3	0.573
…	…
433	0.505
434	1.000
435	0.722

$dist_{basic}$ sorted in descending power

Index	Distance	RANK/435
434	1.000	0.002
322	0.999	0.005
280	0.997	0.007
…	…	…
73	**0.040**	**0.95**
…	…	…
169	**0.012**	**0.99**
…	…	…
364	0.009	0.996
51	**0.007**	**0.998**
422	0.001	1.000

Conceptual image of threshold T (Y= RANK/435, X=Distance)

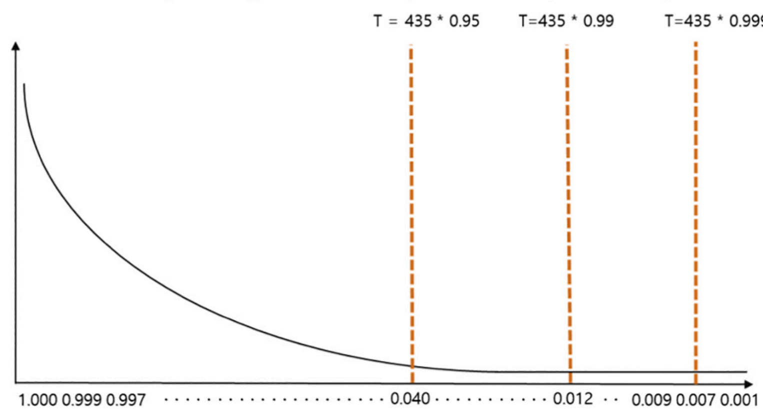

Fig. 3. Illustrative application of Algorithm 2.

Algorithm 3 Distance matrix corresponding to a basic database and an extended database

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Input: A basic database, DB_{basic}, and an extended database, $DB_{extended}$, which share critical tuples with each other

Output: Distance matrix, $dist_{extended}$, between a basic database and an extended database

for all $i = 1$ to n **do** // n equals the number of rows in DB_{basic}
 for all $j = 1$ to m **do** // m equals the number of rows in $DB_{extended}$
  if $dist_{extended}$ is taken to be Euclidean distance **then**
   return $dist_{extended} = D_{euclidean}(DB_{basic}[i], DB_{extended}[j])$
  else $dist_{extended}$ is taken to be Mahalanobis distance **then**
   return $dist_{extended} = D_{mahalanobis}(DB_{basic}[i], DB_{extended}[j])$
  end if
 end for
end for
return $dist_{extended}$

Algorithm 4 Determination of success or failure of linkage

Input: The distance matrix, $dist_{extended}$, corresponding to a basic database and an extended database and the value of the threshold to be used for record linkage, T

Output: Whether the entries of $dist_{extended}$ satisfy the condition given by the threshold T

1: **sort in descending power** distance distribution of $dist_{extended}$

2: **if** an entries of $dist_{extended}$ is equal to or less than threshold T **then**

3: **return** record linkage success between the corresponding records

4: **else** $dist_{extended}$ is greater than the threshold T **then**

5: **return** record linkage failure between the corresponding records

6: **end if**

7: **return** Whether the entries of $dist_{extended}$ satisfy the condition given by the threshold T

Algorithm 5 Micro-aggregation to generate synthetic data

Input: The set of records that are successfully linked to each record in the basic database, $recordList$, and the minimum number of records for micro-aggregation, k (greater than or equal to 3)

Output: Synthetic data generated by micro-aggregation based on the minimum number, k, of entries in $recordList$

1: **set** the number of records needed for micro-aggregation as parameter k

2: **for all** $i = 1$ to n **do** // n equals the number of rows in DB_{basic}

3: **compute** Average of records in $recordList$ that share links with k or more individual records in DB_{basic}

4: **end for**

5: **return** Synthetic data generated by micro-aggregation

distance exhibits a wider record linkage list pool and a lower hit rate, whereas the Mahalanobis distance exhibits a narrower record linkage list pool but a higher hit rate. Further, with an increase in the confidence level, both distance metrics exhibited decrements in the corresponding hit rates.

4.2.3. Evaluation of data privacy levels

We conducted micro-aggregation via Algorithm 5 in order to evaluate the privacy disclosure risks and generate appropriate synthetic data. Here, the average of the distances corresponding to records in the extended database that correspond to 3 or more linkages were calculated. Using the obtained results, the risk of the original data and the risk of the synthetic data generated via the proposed method were compared to assess the privacy disclosure risk of the proposed method.

The results of the comparison are as follows. With respect to global

Table 1
Comparison of the overall performances of the models.[a]

Hyperparameter		Linkage performance		Utility performance	Privacy protection performance
Distance	Threshold level	Basic Database	Extended Database	Hit rate	individual risk RMSE
EUC	95%	67.8%	43.65 cases	0.423	0.102***
EUC	99%	67.6%	21.63 cases	0.402	0.015
EUC	99.9%	61.9%	3.79 cases	0.297	0
MAH	95%	86.2%	10.03 cases	0.485	0.138
MAH	99%	84.6%	5.59 cases	0.429	0.135
MAH	99.9%	68.6%	1.68 cases	0.343	0.113

$p < .001$ ***, minimum number of records for micro-aggregation = 3.

[a] Linkage performance refers to the result of linking records. Utility performance refers to the hit rate of a record (true label) that shares an actual identifier with the linked records. Privacy protection performance refers to RMSE, which represents the difference between the results of the *k*-anonymity test between synthetic data and original data generated by micro-aggregation of at least three linked records. Finally, the statistical difference was verified via *t*-test of the two data corresponding to this RMSE. In all hyperparameter combinations, synthetic data did not impede the degree of de-identification of the original data.

risk[2] of distance parameter and threshold level parameter combinations, the synthetic data exhibited better overall privacy protection performance than the original data. In particular, this was the case in the majority of the instances when the re-identification risk was measured via the *k*-anonymity test. Moreover, the results of statistical evaluation on the difference in risk corresponding to each record in the basic and synthetic datasets were as follows. The RMSE values of individual risks[3] were lower in the synthetic database based on the Euclidean distance than the synthetic database based on the Mahalanobis distance. The actual statistical significance of the difference between these values was evaluated via paired *t*-tests using the individual risks of the original data and the synthetic data. No statistically significant difference was observed except the EUC_{95} case, which is combination of parameters with Euclidean distance and a threshold of 0.95 ($p > .05$). Meanwhile, in the case of EUC_{95}, individual risks of the synthetic data were observed to be significantly lower than those of the original data. Table C in the appendix presents a summary of these results.

This completes a thorough evaluation of the performance of the proposed model in terms of various evaluation indices such as record linkage, data utility, and privacy protection. The results of these evaluations have been summarized in Table 1. Linkage performance of the basic database is represented by $\frac{\text{number of record linkage successes}}{\text{total number of records in the basic database}}$. Similarly, linkage performance of the extended database is represented by the average number of entries in the record linkage list per basic database record. As is evident from the data, the performance of the proposed method depends on the combinations of distance metrics used and the threshold parameters selected.

5. Discussion

The evaluation was framed with the objective of assessing the balance achieved by the optimal generated synthetic data between data utility and privacy. The combination of Mahalanobis distance and a privacy threshold level of 99% (MAH 99%) is ascertained to be the optimal parameter configuration in this respect.

In this study, the combination with a small number of records linkage cases to the extended database and with the highest as possible hit rate was considered the optimal parameter combination. This is because the above combination of parameters has less distortion of the original data in the basic database and can produce the synthetic data that best preserves the statistics of the original data from the basic database. Based on this concept, MAH 99% was evaluated as the optimal parameter combination. The detailed description is as follows.

The method proposed in this study can be divided into a linkage process based on comparison between calculated distances and a predefined threshold and a de-identification process based on micro-aggregation. At the first part, the linkage process could include a case involving no linkages in the extended database corresponding to individual records in the basic database owing to unfulfilled threshold conditions. Because of this issue, the linkage performance of the basic database, one of the performance evaluations we proposed, means the ratio excluding records of the basic database, which has no linked results, as described earlier in the total number of records in the basic database. Conversely, the linkage performance on the extended database is represented by the average of the number of records in the extended database linked to individual records in the basic database.

When Euclidean distance was used as a metric for record linkage, the model exhibited a low linkage rate in the basic database, which was used as a reference, but exhibited a wide record linkage list pool in the extended database. Further, the data utility achieved was relatively low, whereas the privacy protection level was relatively high.

By contrast, utilization of Mahalanobis distance yielded relatively high linkage rate in the basic database, which is the reference that researchers are interested in, but the record linkage list pool in the extended database was relatively small. Further, in this case, the model exhibited high data utility and low privacy protection level. According to the results of the evaluation of distance metrics, Mahalanobis distance was rated to perform better than Euclidean distance because of its high success linkage rate in basic databases and the low number of linkage case in extended databases.

In addition, the evaluation indicated that the privacy threshold level performs fine-tuning of the proposed model. Increasing the privacy threshold level slightly reduces the data utility, but significantly reduces the number of records linkage with an extended database from the basic database. If you want to give a lot of distortion to the original data in the basic database when generating synthetic data, set the privacy threshold level low. Conversely, if you want to give less distortion to the original data in the basic database, set the privacy threshold level high.

Through the above process, new synthetic data that preserved the statistical characteristics of the original data was generated via micro-aggregation that converted the original data to the average of the records linked by the de-identification process. A comparison between the privacy disclosure risks of the generated synthetic data and the original data revealed no case in which the synthetic data hindered the degree of de-identification of the original data corresponding to all hyperparameter combinations as mentioned above. Therefore, it can be concluded that the proposed method does not suffer from any risk of data re-identification.

Corresponding to MAH 99%, the linkage performance of the basic database was observed to be 84.6% and the linkage performance of the extended database was observed to be 5.59 on average. This is a reasonable number of linked records in the case of most basic databases. Meanwhile, the hit rate, which was used to capture whether the linked records contained a record that shared the actual identifier, was observed to be 0.429. This is higher than the hit rate of 0.423 corresponding to the combination of Euclidean distance and a privacy threshold level of 95% (EUC 95%), which exhibited an average of 43.65 linked records, and can be regarded to correspond to relatively very

[2] Global disclosure risk is defined in terms of the expected number of identifications in the released microdata. It measures the proportion of correct linkages amongst those records in the population that link a sample unique masked microdata record (Elliot, 2000; Skinner & Elliot, 2002).

[3] The risk is computed for every released entity from the masked microdata. In this scenario, the individual risk for each entity is given by the probability of correct identification by an intruder (Truta, Fotouhi, & Barth-Jones, 2006).

good performance.

The proposed method was inspired by the works of Domingo-Ferrer and González-Nicolás (2010). However, the method proposed in this study is considerably different as that it can be used on data without identifiers. In addition, it can be applied to heterogeneous data mining via record linkage that converts all records into new values that preserve the statistical characteristics of the original data. While the method developed by Domingo-Ferrer and González-Nicolás transforms the original data to lower the privacy disclosure risk of a single database with an identifier, the method proposed in this study combines the transformed heterogeneous data with the original data to construct an integrated database, enabling the mining of two or more databases without identifiers.

6. Conclusions

As a result of the G8 Charter, announced in 2013, many countries are currently participating in the activation of OGD. However, a series of practical difficulties are observed in the context of data mining OGD for in-depth analysis. First, identifiers required to prevent privacy disclosure are absent in almost all OGDs. Second, owing to data silos, the data sharing and collection methods differ between heterogeneous OGDs, and administrative or institutional barriers need to be overcome. This has created a demand for a technical solution to these issues. To address this, a new algorithm based on distance-based record linkage and micro-aggregation was proposed in this study to strike a balance between data utility and privacy disclosure during the data mining of sensitive information without any identifiers. The model was observed to exhibit satisfactory results. The most fundamental feature of the proposed model is its utilization of hyperparameters, such as distance and threshold, to adequately control the balance between data utility and privacy disclosure. Based on extensive experimentation, the relationship between data utility and privacy disclosure levels and various hyper-parameter combinations was derived. Subsequently, an optimal parameter configuration was identified. The academic contributions of this study are as follows. First, the majority of the previous studies have considered the problem of privacy preservation during the publication stage of the OGD. However, this study proposed a method to preserve and utilize the privacy of already published OGD. In this process, it emphasized the importance of research on already published OGD that has, thus far, been relatively neglected. Further, it has broadened the research area for PPDM by proposing a method to create an integrated dataset for the mining of heterogeneous data without any identifiers. We intend to investigate methods to train the proposed model in future works to improve its practical utility by making it faster and easier to reuse. In addition, a guide will be provided to policy makers for reference by conducting a case study using the proposed method.

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Appendices

Table A
Status of privacy protection legislations in major countries.

Category	Anonymization (or de-identification)-based legislation			Mixed legislation, comprising anonymization and pseudonymization
	EU	USA	Korea	EU
Law	DP Directive[a]	HIPAA[b]	Personal Information Protection Act[c]	GDPR[d]
Law Clause	(Recital26) Anonymous Information Reference: exclusion of protection measures Delegated to EU members as code of conduct	De-identified medical information does not apply to the law	(Article 3) Reference to anonymization included in Section 7 of the Personal Information Protection Principles (Article 18) Restrictions on the use and provision of personal information for purposes other than the intended one applies to exceptions provided Section 2	(Recital 26) Anonymized information is not subject to GDPR. Pseudonymized information is regarded to be personal information. (Article 4) Definition of Pseudonymization (Article 89) For public interest, scientific or historical research purposes or statistical purposes, pseudonymization may be applicable.(Permitted to third parties)
Code of Conduct	(2014.4) WP29 opinion on Anonymization Techniques	(2002) HIPAA Privacy Rule – De-identification (2015) NISTIR De-identification of Personal Information	(2016.6) Guidelines for de-identification of personal information	(2014.4) WP29 opinion on Anonymization Techniques

Source: (Kim, 2017).
[a] Directive 95/46/EC of the European Parliament and the Council of 1995.
[b] Health Insurance Portability and Accountability Act of the US Department of Health and Human Services (HHS) of 1996.
[c] Personal Information Protection Act of the Korea Ministry of the Interior and Safety of 2011.
[d] Regulation (EU) 2016/679 of the European Parliament and of the Council of 2016.

Table B
Details of de-identification techniques.

Processing Technique	Category	Description	Integrated Technology
Pseudonymization	Concept Processing Target Advantages	Replaces personally identifiable data with other values that are not directly identifiable Name, other unique identifiers (Zip code, etc.)	-Heuristic Pseudonymization -Encryption -Swapping

(continued on next page)

Table B (*continued*)

Processing Technique	Category	Description	Integrated Technology
		Possible to completely de-identify data Creates little to no data transformation and deterioration	
	Disadvantages	Limits name-based analysis due to the pseudonymization of generalized substitution values	
Aggregation	Concept	Applies statistical values to personal information to make it impossible to identify specific individuals	-General Aggregation
	Processing Target	Information related to individuals (name, etc.), other unique features (income, expenditure, etc.)	-Micro-aggregation -Rounding
	Advantages	De-identifies sensitive information Useful to generate various datasets for statistical analysis	-Rearrangement
	Disadvantages	Difficult to perform precise analysis based on aggregated data Possible to extract or predict personal information during data combining process if the amount of aggregated data is small	
Data Reduction	Concept	Deletes specific data values that can be used to identify personal information	-Reducing Records
	Processing Target	Unique identification number (social security number, etc.)	-Reducing Identifiers
	Advantages	Prevents predictions and inferences via complete deletion of sensitive personal information	
	Disadvantages	Reduced diversity, effectiveness, and reliability of analysis results due to complete data deletion	
Data Suppression	Concept	Prevents unique information tracking and identification by converting the given identification information to a representative value of groups or to pre-defined ranges	-Random rounding -Controlled rounding
	Processing Target	Unique identification number (social security number, etc.)	-Data Range
	Advantages	Enables various analysis and processing due to its format of statistical data	
	Disadvantages	Difficult to derive analysis results for exact numerical values	
Data Masking	Concept	Converts personal identification information into alternate values such as blank spaces and '*' noise	-Adding Random Noise
	Processing Target	Unique identification number (social security number, etc.)	-Blank & Impute
	Advantages	Possible to conduct complete de-identification Creates fewer raw data structural transformations	
	Disadvantages	Various limitations may be induced during the utilization of necessary information if excessive masking is performed	

Source: Korea Ministry of the Interior and Safety (2016).

Table C
Comparison of the privacy protection performances of the models.

Parameter combination		Global risks	k-anonymity test			Individual risk	
			k = 2	k = 3	k = 5	RMSE	t-test
EUC95	Origin	0.269	0.092	0.196	0.377		
(*n* = 2319)	Synthetic	0.252	0.088	0.182	0.336	0.102	0.028***
EUC99	Origin	0.263	0.084	0.189	0.372		
(*n* = 2285)	Synthetic	0.262	0.083	0.189	0.372	0.015	0.954
EUC999	Origin	0.279	0.089	0.196	0.389		
(*n* = 1561)	Synthetic	0.279	0.089	0.196	0.389	0	1
MAH95	Origin	0.221	0.069	0.152	0.292		
(*n* = 2944)	Synthetic	0.218	0.069	0.144	0.285	0.138	0.713
MAH99	Origin	0.221	0.063	0.149	0.299		
(*n* = 2691)	Synthetic	0.221	0.068	0.141	0.304	0.135	0.955
MAH999	Origin	0.297	0.108	0.205	0.458		
(*n* = 1154)	Synthetic	0.296	0.109	0.215	0.42	0.113	0.94

$p < .001$ ***, minimum number of records for micro-aggregation = 3.

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